The Devil's Dung? Money as a mechanism of generalized reciprocity in human societies

Supporting Information

Section 1 provides the complete model pseudocode, while Section 2 presents the flowchart.

Section 3 includes sensitivity analysis and robustness tests.

1 Model Pseudocode

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In an agent-based environment,
1. Setup:
        1.1 Define global parameters:
                set BENEFIT-TO-COST-RATIO [variable input > 1]
                set COST 1
                determine BENEFIT [BENEFIT-TO-COST-RATIO * COST]
                set INITIAL-REPUTATION [variable input, 1 by default]
                set INITIAL-LIQUIDITY [variable input > 0]
                set COOPERATION-COUNTER 0
        1.2 Create agent population:
                1.2.1 Define agent types, create a given number of agents for each type:
                        Create Unconditional Cooperators [variable input, 100 by default]
                        Create Defectors [variable input, 100 by default]
                        Create Direct-Reciprocators [variable input, 100 by default]
                        Create Indirect-Reciprocators [variable input, 100 by default]
                        Create Money-Users [variable input, 100 by default]
                1.2.2 Define initial agent variables. For every agent:
                        set ID [unique identifier]
                        set fitness 0
                        set current-partner NA [ID of another agent]
                        set memory [empty list]
                        set reputation-score [INITIAL-REPUTATION, 1 by default]
                        set money-balance [result of assign-initial-money function]
        1.3 Define auxiliary functions:
                function help-partner():
                        decrease acting-agent's fitness by COST
                        increase current-partner's fitness by BENEFIT
                        increase acting-agent's reputation-score by 1
                        increase COOPERATION-COUNTER by 1
                function do-not-help():
                        add acting-agent to current-partner's memory
                        decrease acting-agent's reputation-score by 1
                function assign-initial-money():
                        if INITIAL-LIQUIDITY < 1:
                                if random uniform (0,1) < INITIAL-LIQUIDITY:
                                        set money-balance 1
                                else:
                                        set money-balance 0
                        else:
                                set money-balance INITIAL-LIQUIDITY
2. Dynamics: at every time step, iterate through all agents twice:
        2.1 First iteration: for each agent,
                set fitness 0
                set current-partner = a randomly selected other agent
        2.2 Second iteration: for each agent,
                if type = "Unconditional Cooperator":
                        help_partner()
                if type = "Defector":
                        do_not_help()
                if type = "Direct-Reciprocator":
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                if current-partner is not in memory:
                        help_partner()
                else:
                        do_not_help()
                        remove current-partner from memory
        if type = "Indirect-Reciprocator":
                if reputation-score of current-partner > 0:
                        help_partner()
                else:
                        do_not_help()
        if type = "Money-User":
                if money-balance of current-partner > 0:
                        help_partner()
                        increase acting-agent's money-balance by 1
                        decrease current-partner's money-balance by 1
                else:
                        do_not_help()
3. Strategy evolutionary updating: After all agents have interacted, a random agent is
selected to potentially change its strategy. This agent may adopt a new strategy based on
the success of other surviving strategies in the current round, as in a roulette wheel
selection process. The probability of adopting a specific strategy is weighted according to
its relative in-round fitness. To avoid negative probabilities, each agent's fitness is
increased by one unit at the end of each time step.
4. Collect metrics: At the end of each iteration, collect data on the number of agents
following each strategy and the total cooperation rate observed in the population. After
data collection, reset COOPERATION-COUNTER to 0.
5. Record metrics: Every 250 iterations, log the number of agents following each strategy
and the overall cooperation rate in the population to a .csv file.
The simulation is interrupted after a predetermined number of time steps (30000 by default).
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2 Model Flowchart

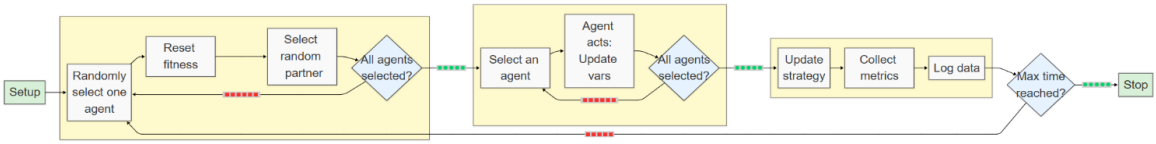

Figure S1. Model execution flowchart. Red squares indicate "No", while green circles indicate "Yes" decisions at branching points.

3 Sensitivity Analysis

Figure S2 shows the results of the evolutionary tournament in the benchmark scenario, where agents following the monetary exchange strategy were removed from the initial population. The population size was fixed at $N = 500$, with an equal distribution of strategies: 125 unconditional cooperators, 125 direct reciprocators, 125 indirect reciprocators, and 125 defectors.

Given the settings of our tournament, a combination of the three cooperating strategies (unconditional cooperators, direct reciprocators, and indirect reciprocators) can outcompete and eliminate defectors, but only at relatively high benefit-to-cost ratios. Note that Figure 2 of the main manuscript highlights the broader evolutionary advantage conferred by money-mediated cooperation circuits (for a simplified visualization of the formation of such circuits, see Figure 6 of the main manuscript). Indeed, the inclusion of the monetary exchange strategy

allows for the exclusion of defectors even at much lower benefit-to-cost ratios (as low as 1.25), provided that initial liquidity is not excessive.

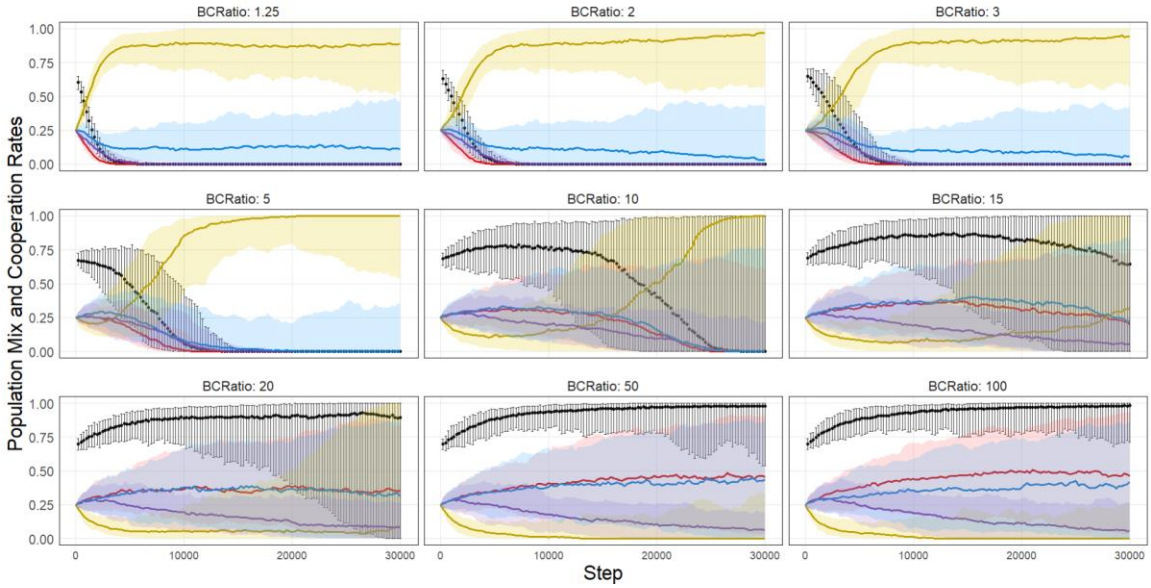

Figure S2. Evolutionary dynamics of population composition and cooperation in the benchmark tournament model under selected benefit-to-cost ratios. In this scenario, agents using the monetary exchange strategy were removed from the initial population, and the prevalence of the remaining four strategies was scaled proportionally. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQRs). Specifically, yellow denotes defectors, red unconditional cooperators, light blue indirect reciprocators, and purple direct reciprocators. The black line and associated error bars show the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

It is worth reiterating that, when defectors are effectively eliminated at sufficiently high benefit-to-cost ratios, reciprocity-based strategies alone are unable to eliminate unconditional cooperators from the population (Figure S2). As discussed in the main manuscript, unconditional cooperators can be viewed as *neutral mutants* of reciprocity strategies, leaving the system vulnerable to late invasions by defectors or mutations that promote defection. In contrast, the monetary exchange strategy successfully excludes unconditional cooperators (see Figure 5 of the main manuscript), thereby increasing the system's resilience to late-stage invasions by defectors.

Figure S3 illustrates these dynamics in detail. For simplicity, and without loss of generality, we assume the same benefit-to-cost ratios and liquidity levels as in Figure 2 of the main manuscript. The tournament setup is also identical to Figure 2, with one key difference: at step 15000, we

forcibly convert half of the agents following the monetary exchange strategy into randomly selected defectors.

By the time this mass invasion occurs, the monetary exchange strategy has already become an evolutionarily stable under all parameter combinations shown in Figure S3, and cooperation rates have stabilized at levels that vary with initial liquidity (see also Figure 3 of the main manuscript). Since agents are randomly converted, there is an equal probability of selecting individuals with high or low balances. This results in a variable number of defectors with high balances, which, as discussed in the main manuscript, poses a threat to the stability of money-mediated cooperative circuits.

Nevertheless, we observe that after this shock, defectors are once again negatively selected in favor of the monetary exchange strategy. Cooperation rates quickly return to their pre-invasion equilibrium levels. This result underscores the robustness of the monetary exchange strategy against late invasions by defectors.

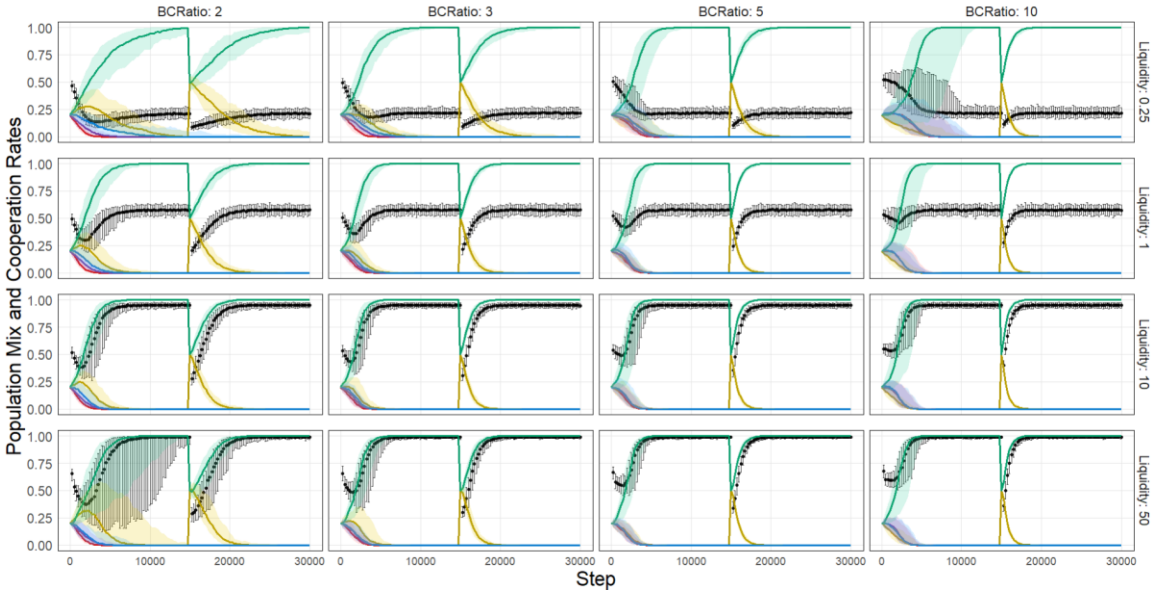

Figure S3. Robustness of the monetary exchange strategy to a mass invasion of defectors. This figure replicates the tournament setup from Figure 2 of the main manuscript, with one important difference: at the 15000th simulation step, half of the agents following the monetary exchange strategy were forcibly converted into defectors at random. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQRs). Specifically, green indicates agents following the monetary exchange strategy, yellow indicates defectors, red indicates unconditional cooperators, light blue indicates indirect reciprocators, and purple indicates direct reciprocators. The black line and associated error bands indicate the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps. Across all conditions, the monetary strategy consistently regains dominance after

the invasion, and cooperation rates begin to recover towards the same equilibrium levels as in Figure 2 of the main manuscript.

Figures S4 and S5 provide additional tests of the robustness of the monetary exchange strategy by examining its stability when forcibly converting half of the agents into either unconditional cooperators (Figure S4) or a mix of unconditional cooperators and defectors (Figure S5). Consistent with the results in Figure 5 of the main manuscript, Figure S4 shows that money is not a neutral mutant of unconditional cooperators, as they are rapidly eliminated from the population after the shock.

Interestingly, a mass invasion of unconditional cooperators has opposite short-term effects on cooperation at the population level, depending on the level of liquidity. When liquidity is low (e.g., one token or less per agent), cooperation initially increases; when liquidity is sufficiently high, it decreases. This pattern arises because forcibly converted unconditional cooperators always help agents in the monetary exchange circuit, while these agents only reciprocate if the cooperators have tokens. After conversion, several unconditional cooperators may start with zero token balances and, by design, do not accumulate tokens over time. As a result, the average level of cooperation is lower than in a fully monetary system where tokens circulate dynamically. However, under conditions of low liquidity conditions—where cooperation within the monetary circuit is already limited (see Figure 3 of the main manuscript)—the additional cooperative acts performed by unconditional cooperators temporarily increase overall cooperation. Over time, as these cooperators exhaust their tokens through repeated interactions, they cease to receive cooperation from monetary agents and are effectively excluded from the circuit. As a result, they are gradually negatively selected, and the system returns to its equilibrium level of cooperation.

Figure S5 shows that converting half of the agents following the monetary exchange strategy into a mix of defectors and unconditional cooperators has no lasting effect on the stability of the system. In this scenario, the forced mutation introduces defectors who benefit from the unconditional help of cooperators, making defection temporarily even more fit than in the case shown in Figure S3. However, this advantage is not sufficient to destabilize the money-induced cooperation circuit, which remains resilient to the shock.

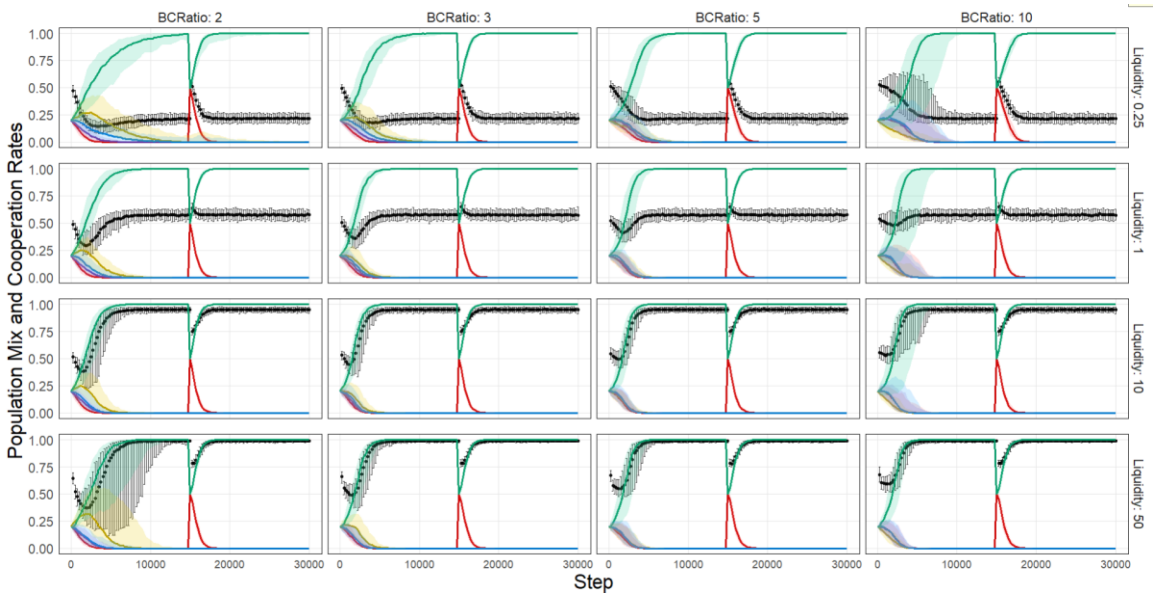

Figure S4. Robustness of the monetary exchange strategy to a mass invasion of unconditional cooperators. This figure replicates the tournament setup from Figure 2 of the main manuscript, with one key difference: at the 15000th simulation step, half of the agents following the monetary exchange strategy were forcibly converted into unconditional cooperators at random. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQRs). Specifically, green indicates agents following the monetary exchange strategy, yellow indicates defectors, red indicates unconditional cooperators, light blue indicates indirect reciprocators, and purple indicates direct reciprocators. The black line and associated error bands indicate the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps. Across all conditions, the monetary strategy consistently regains dominance after the invasion.

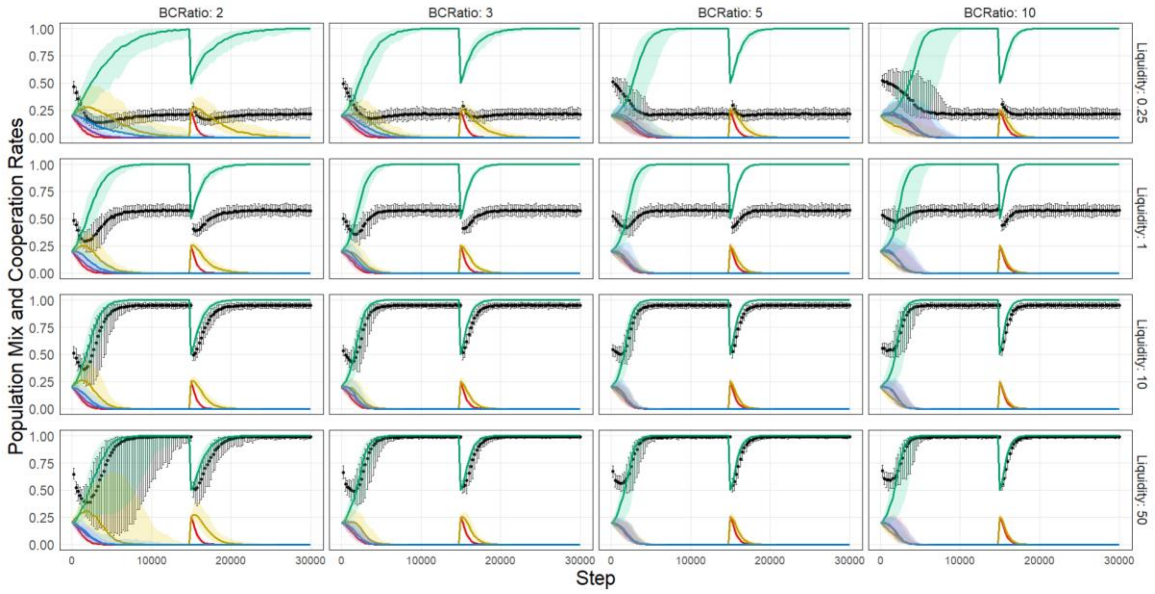

Figure S5. Robustness of the monetary exchange strategy to a mass invasion of a mix of unconditional cooperators and defectors. This figure replicates the tournament setup from Figure 2 of the main manuscript, with one key difference: at the 15000th simulation step, half of the agents

following the monetary exchange strategy were forcibly converted into either unconditional cooperators or defectors at random. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQRs). Specifically, green indicates agents following the monetary exchange strategy, yellow indicates defectors, red indicates unconditional cooperators, light blue indicates indirect reciprocators, and purple indicates direct reciprocators. The black line and associated error bands indicate the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps. Across all conditions, the monetary strategy consistently regains dominance after the invasion.

Figures S6 and S7 illustrate the evolutionary trajectories and cooperation rates in populations initially composed exclusively of unconditional cooperators, defectors, and agents adopting the monetary exchange strategy. In Figure S6, the simulation starts with 167 agents per strategy (based on rounding up $N = 500$ divided by three), while in Figure S7 the initial population is increased by 20% to $N = 600$, with 200 agents per strategy. Across all scenarios examined, the monetary exchange strategy consistently outperforms the other two, ultimately emerging as the evolutionarily stable strategy. This dominance is driven by the formation of money-mediated cooperation circuits that systematically exclude both defectors and unconditional cooperators from participating in cooperative exchanges.

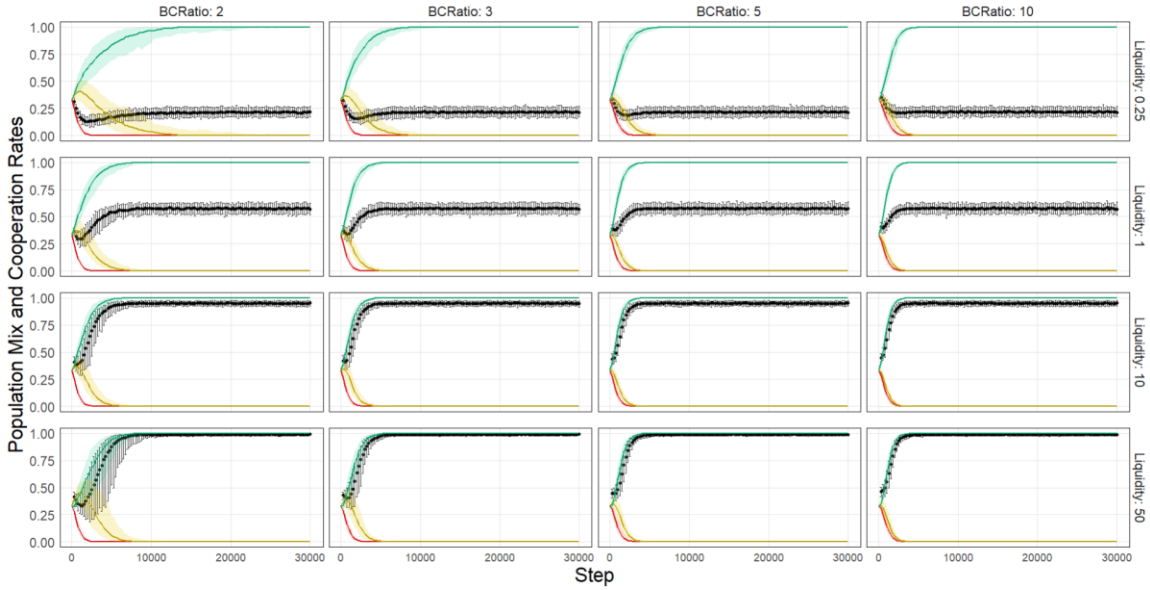

Figure S6. Evolutionary dynamics of population composition and cooperation in the tournament model under selected benefit-to-cost ratios. In this scenario, the initial population consists exclusively of unconditional cooperators (red), defectors (yellow), and agents following the monetary exchange strategy (green), with reciprocity strategies removed. The simulation starts with 167 agents per strategy. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQR). The black line and associated error bars show the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

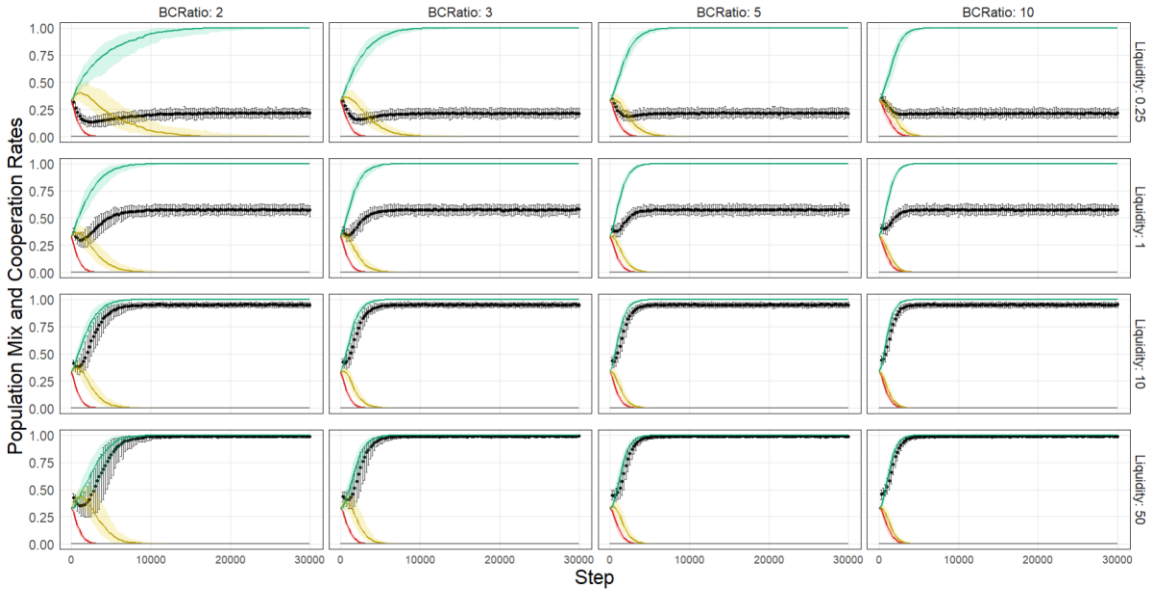

Figure S7. Evolutionary dynamics of population composition and cooperation in the tournament model under selected benefit-to-cost ratios. In this scenario, the initial population consists exclusively of unconditional cooperators (red), defectors (yellow), and agents following the monetary exchange strategy (green), with reciprocity strategies removed. The simulation starts with 200 agents per strategy. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQR). The black line and associated error bars show the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

Figures S8 and S9 replicate Figure 4 of the main manuscript, showing the equilibrium cooperation rates and the dominant strategy across all 117 combinations of liquidity and benefit-to-cost ratios. These simulations assume initial populations consisting of either 167 (Figure S8) or 200 (Figure S9) unconditional cooperators, defectors, and agents following the monetary exchange strategy. The patterns observed in Figures S8 and S9 closely mirror those in Figure 4 of the main text. In particular, we again observe the excess liquidity effect i.e., a decrease in cooperation due to the proliferation of defectors at very high liquidity levels when the returns to cooperation are low. For the remaining parameter combinations, the monetary exchange strategy remains dominant, with cooperation rates generally increasing with liquidity.

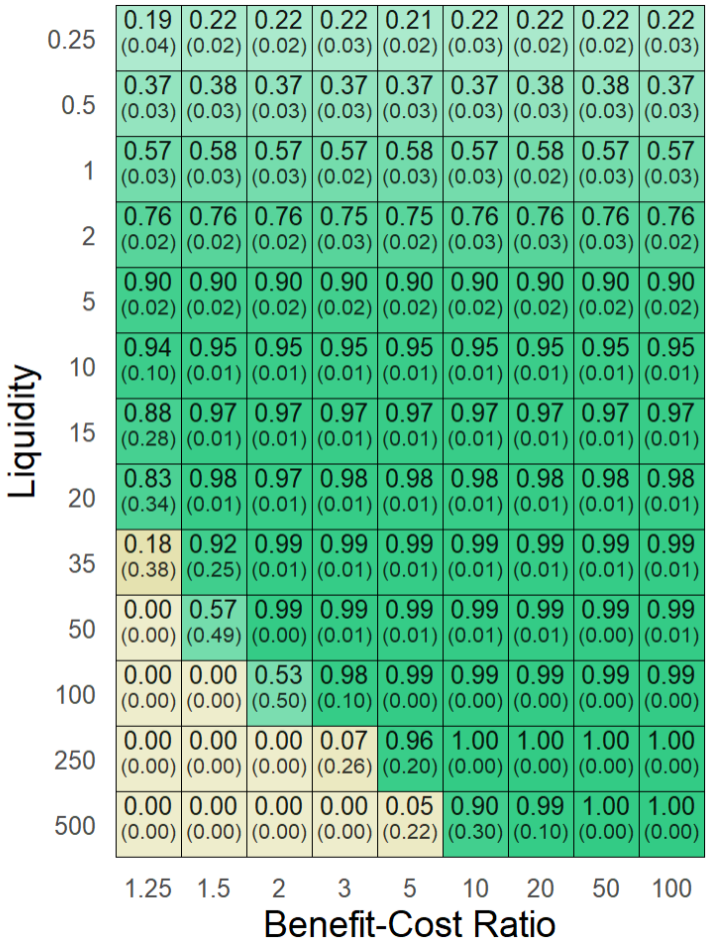

Figure S8. Cooperation rates and dominant strategies at the end of the simulation (30000th step) across 117 combinations of benefit-to-cost ratios and liquidity values. Reciprocity strategies are excluded in this scenario. The initial population consists of 167 unconditional cooperators, 167 defectors, and 167 agents following the monetary exchange strategy. Larger numbers indicate the mean cooperation rate for each combination of benefit-to-cost ratio and liquidity level, while smaller numbers in parentheses indicating the standard deviations over 100 independent runs. Cell colors represent the dominant strategy: green for money and yellow for defection. A transparency gradient visually illustrates mean cooperation rates across the parameter space, with darker shades indicating higher cooperation.

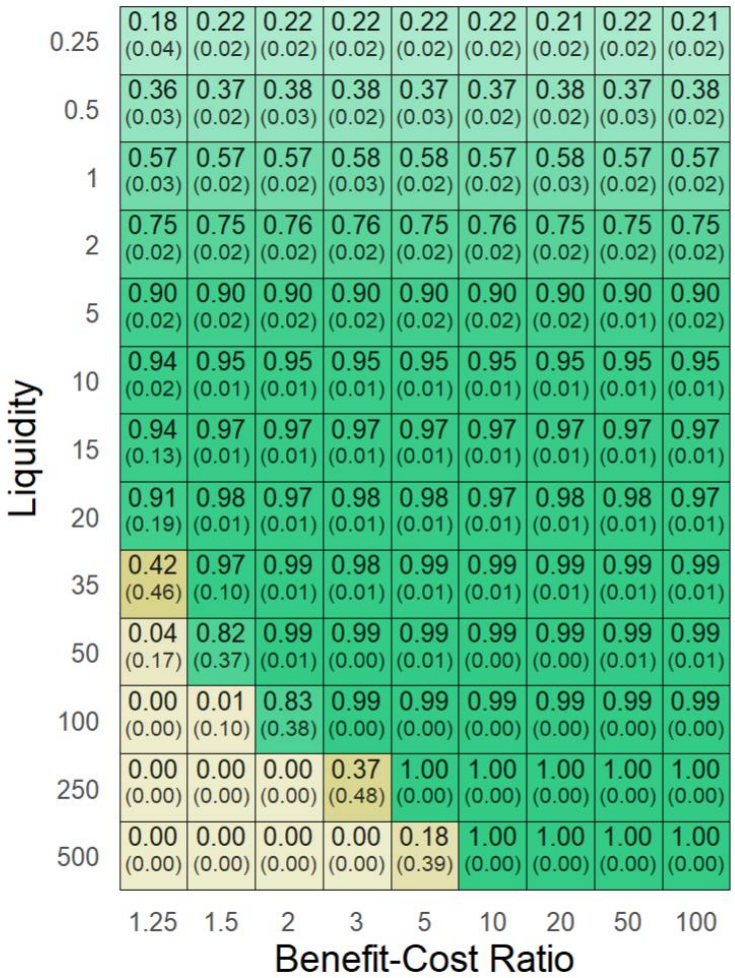

Figure S9. Cooperation rates and dominant strategies at the end of the simulation (30000th step) across 117 combinations of benefit-to-cost ratios and liquidity values. Reciprocity strategies are excluded in this scenario. The initial population consists of 200 unconditional cooperators, 200 defectors, and 200 agents following the monetary exchange strategy. Larger numbers indicate the mean cooperation rate for each combination of benefit-to-cost ratio and liquidity level, with smaller numbers in parentheses indicating the standard deviations over 100 independent runs. Cell colors represent the dominant strategy: green for money and yellow for defection. A transparency gradient visually illustrates mean cooperation rates across the parameter space, with darker shades indicating higher cooperation.

Finally, we present a sensitivity analysis of Figure 2 from the main manuscript focusing on larger initial populations. Here, we examine the effects of a higher initial prevalence of unconditional cooperators (Figure S10), defectors (Figure S11), and a proportionally larger overall population size (Figure S12). In Figure S10, the simulation starts with 400 unconditional cooperators, 100 defectors, 100 direct and indirect reciprocators, and 100 agents following the monetary exchange strategy, for a total population of $N = 800$. In Figure S11, we increase the initial number of defectors to 400 while keeping the total population size at 800 (i.e., we reduce the

number of unconditional cooperators to 100). In Figure S12, the population is expanded, starting with 200 agents for each strategy, for a total of $N = 1000$.

Overall, we observe consistent patterns with those reported in the main manuscript, even when the tournament starts with different initial population compositions and sizes. This highlights the robustness and evolutionary advantage of the monetary exchange strategy. In Figure S10, the initial rates of cooperation are significantly higher than those in Figure 2 of the main manuscript — especially under low liquidity conditions — due to the larger initial proportion of unconditional cooperators. However, as the monetary strategy gains dominance, the cooperation rates converge to the same equilibrium levels.

Perhaps more interestingly, Figure S11 shows that a higher initial proportion of defectors (50% of the population) amplifies the excess liquidity effect, especially when the returns to cooperation are relatively low. The first signs of this effect already appear at a liquidity level of 50 and become clearly pronounced at liquidity = 100. Note that we have included this additional liquidity value in Figure S11 to specifically illustrate the earlier onset of the excess liquidity effect under these conditions.

As shown in Figure 4 of the main manuscript and in Figures S8 and S9, when the population is initially balanced, the excess liquidity effect at liquidity = 100 typically emerges only for benefit-to-cost ratios below 2. In contrast, Figure S11 shows that within an initially defector-dominated population, this destabilizing effect can emerge even when the benefit-to-cost ratio is as high as 3. These results reinforce the idea that the presence of a significant number of defectors with sufficient money balances can severely undermine cooperation, especially under low returns to cooperation, thereby reducing the effectiveness of the monetary mechanism.

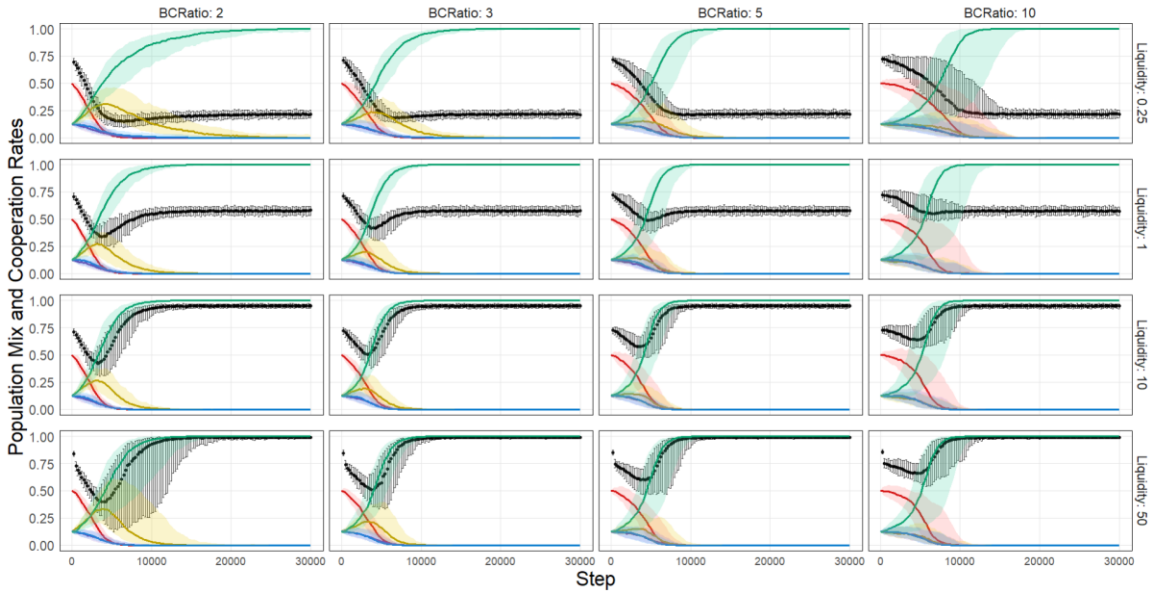

Figure S10. Evolutionary dynamics of population composition and cooperation in the tournament model under selected benefit-to-cost ratios. The initial population has 800 agents: 400 unconditional cooperators, with the remaining 400 agents evenly distributed across the other four strategies (100 agents per strategy). Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQRs). Specifically, green indicates the monetary exchange strategy, yellow indicates defectors, red indicates unconditional cooperators, light blue indicates indirect reciprocators and purple indicates direct reciprocators. The black line and associated error bars show the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

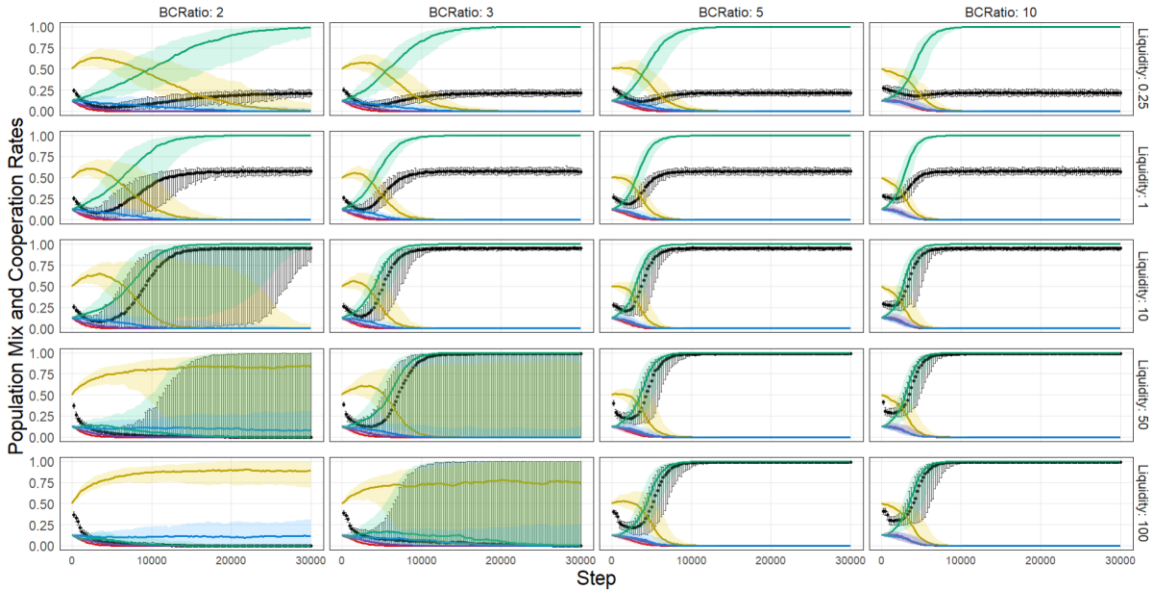

Figure S11. Evolutionary dynamics of population composition and cooperation in the tournament model under selected benefit-to-cost ratios. The initial population consists of 800 agents: 400 defectors, with the remaining 400 agents evenly distributed across the other four strategies (100 agents per strategy). Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQRs). Specifically, green

indicates the monetary exchange strategy, yellow indicates defectors, red indicates unconditional cooperators, light blue indicates indirect reciprocators and purple indicates direct reciprocators. The black line and associated error bars show the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

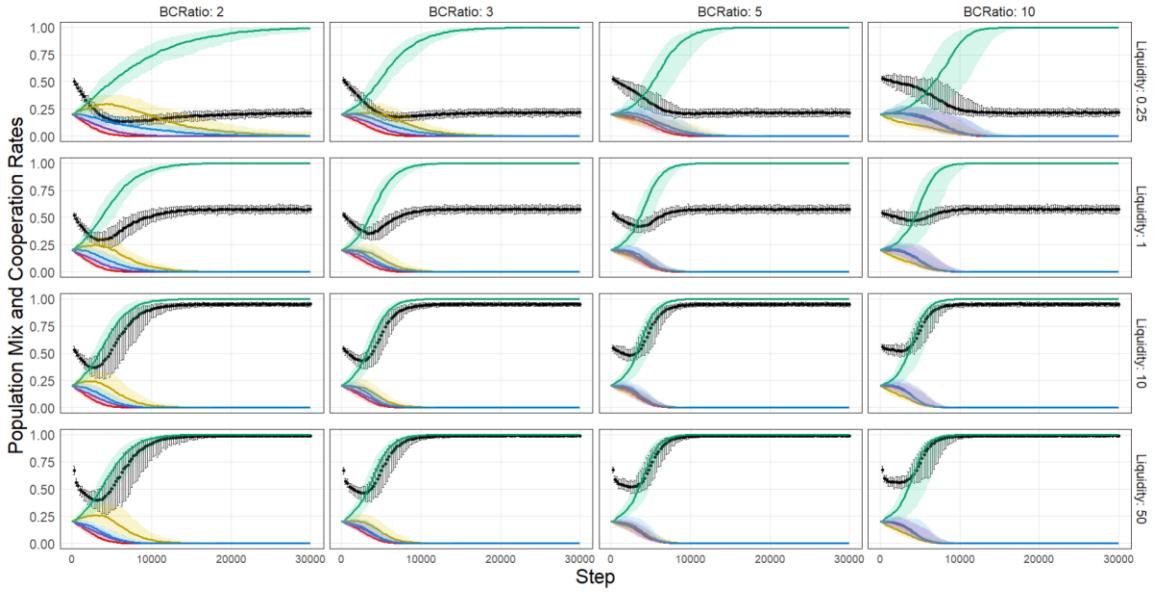

Figure S12. Evolutionary dynamics of population composition and cooperation in the tournament model under selected benefit-to-cost ratios. The initial population comprises 1000 agents, evenly distributed across five strategies (200 agents per strategy). Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQR). Specifically, green indicates the monetary exchange strategy, yellow indicates defectors, red indicates unconditional cooperators, light blue indicates indirect reciprocators and purple indicates direct reciprocators. The black line and associated error bars show the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.